UniGuide AI: A RAG-Based Assistant for Answering University Admissions Questions

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Abstract

This paper describes UniGuide AI, a question answering system built using Retrieval-Augmented Generation (RAG) to help prospective students get accurate answers about University of New Haven (UNH) admissions. The main motivation behind this project is that UNH admissions information is scattered across many different web pages, which makes it hard for students to find what they need quickly. Instead of relying on a language model to remember institution-specific facts which often leads to hallucinated answers this system retrieves relevant text from a small set of official UNH documents and uses that as context for generation. I compared two language models: Gemma-2b-it as the main generator and DistilGPT2 as a simpler baseline. Across ten admissions questions, Gemma-2b-it scored an average of 3.75 out of 5 and won all ten comparisons, while DistilGPT2 averaged 1.0 and failed on every question. The results show that combining retrieval with an instruction-tuned model works well for this kind of domain-specific task, but there are still areas to improve, particularly around how text is chunked and how the retriever handles overlapping topics between documents.

1. Introduction

One thing I noticed as a student at UNH is that finding specific admissions information can be genuinely frustrating. There is no single place that answers common questions like what documents are needed as an international graduate applicant or what the cost of attendance is for students living on campus. You usually end up clicking through multiple pages, and sometimes the information is inconsistent between them. Students who give up searching often end up contacting the wrong office, which wastes time for everyone.

Large language models can answer questions in natural language, but they are not reliable for institution-specific details because they tend to hallucinate generating confident but incorrect information (Lewis et al., 2020). RAG addresses this by first retrieving relevant passages from a document collection and then generating a response conditioned on those passages, so the model is grounded in actual source material rather than relying on what it may or may not have seen during pretraining.

For this project I built a RAG pipeline specifically for UNH admissions Q&A. I collected seven official admissions documents from the UNH website, built a retrieval system using dense embeddings and cosine similarity, and tested two LLM backends: Gemma-2b-it and DistilGPT2. The goal was to see how well a lightweight RAG system can answer real student questions and to understand where it works and where it breaks down.

2. Related Work

The foundational work for this project is Lewis et al. (2020), who introduced the RAG framework and showed that grounding LLM outputs in retrieved passages significantly reduces hallucination on open-domain QA. Their setup uses a large-scale DPR retriever and a seq2seq generator, which is more complex than what I built here, but the core idea is the same.

For the retrieval component, I used the all-MiniLM-L6-v2 model (Reimers & Gurevych, 2019), which is a lightweight sentence encoder designed for semantic similarity. It is practical for small projects because it runs efficiently even on CPU, and it produces good-quality embeddings for domain-specific text when the corpus is small.

The two generation models I used are Gemma-2b-it (Google DeepMind, 2024) and DistilGPT2 (Sanh et al., 2019). Gemma is instruction-tuned, meaning it was trained specifically to follow user instructions, while DistilGPT2 was only trained for next-token prediction with no alignment. The gap in performance between them in this project reflects findings from Wei et al. (2022), who showed that instruction tuning is what enables models to generalize to structured tasks in zero-shot settings. Earlier educational chatbot work like Kazi & Kazi (2017) used simpler keyword matching approaches this project replaces that with dense semantic retrieval and neural generation, which is more flexible but also more complex to tune.

3. System Architecture

The pipeline has five steps: data collection, text chunking, embedding generation, retrieval, and answer generation.

3.1 Data Collection

I manually saved admissions content from the UNH website into seven plain-text files, each covering a different admissions topic: application deadlines, graduate requirements, undergraduate requirements, international student policies, tuition and cost of attendance, FAQ responses, and selected program descriptions. The files were lightly cleaned to remove navigation elements but otherwise kept close to the original source. Total corpus size is around 14,000 tokens.

3.2 Text Chunking

Each document is split into chunks of at most 300 characters. The chunker first tries to split on section headers using a regex pattern, then applies a sliding character window within each section. Chunks shorter than 80 characters are dropped. This produced about 220 chunks total. A limitation here is that 300-character windows sometimes cut sentences in the middle, which means the model occasionally receives incomplete facts for example, a chunk that ends right before a dollar amount that would actually answer the question. This is discussed further in Section 6.

3.3 Embedding and Retrieval

All chunks are encoded with all-MiniLM-L6-v2 into 384-dimensional vectors. At query time, the query is encoded the same way and cosine similarity is computed against all chunk embeddings. I then apply keyword-based adjustments before ranking:

- If query and chunk both mention "deadline": +0.25
- If both mention "international": +0.20
- If query asks about deadlines and chunk is from the deadlines file: +0.30 source boost
- If query asks about deadlines but chunk mentions "spring admission" from a non-deadline file: -0.25 penalty
- Chunks under 80 characters: excluded entirely

These rules were added after observing the retriever pulling undergraduate chunks for graduate-specific queries. The top 5 chunks are concatenated and passed to the generator.

3.4 Answer Generation

The retrieved chunks are placed into a structured prompt that instructs the model to answer in one complete sentence using only the provided context. For Gemma-2b-it I use temperature sampling (temperature=0.3, top_p=0.9, max_new_tokens=100). For DistilGPT2 I use greedy decoding with the same token limit. The prompt is stripped from the output after decoding so only the generated answer is returned.

```
You are a helpful university admissions assistant.
Answer in ONE complete sentence.
If there are multiple deadlines, include all of them with their labels.
Use ONLY exact information from the context. Do not use bullet points.

Context: {retrieved chunks}
Question: {query}
Answer:
```

4. LLM Implementations

4.1 Gemma-2b-it

Gemma-2b-it is a 2.5 billion parameter instruction-tuned causal language model released by Google DeepMind. It is loaded via the Hugging Face transformers library and requires logging in with a Hugging Face token since it is a gated model. I used device_map='auto' so the model layers are automatically placed on the available GPU. It has an 8,192-token context window, which is more than enough for this task. Because it is instruction-tuned, it actually reads and responds to the prompt format this is what makes it effective here.

4.2 DistilGPT2

DistilGPT2 is a distilled version of GPT-2 with 82 million parameters, trained with a standard next-token prediction objective on web text. It has no instruction tuning. I included it as a baseline to test whether retrieval context alone without instruction alignment is enough to produce useful answers. Spoiler: it is not. Its context window is 1,024 tokens, and the pad token was set to the EOS token to avoid warnings during inference.

5. Evaluation

5.1 Evaluation Questions

I developed ten domain-specific questions covering the main topics prospective UNH students ask about. The questions vary from simple single-fact lookups to ones that require combining information from multiple chunks:

- Q1: What is the priority deadline for international graduate students?
- Q2: What are the fall and spring application deadlines for graduate students?
- Q3: What documents are required for a graduate application?
- Q4: Do graduate applicants need GRE or GMAT scores?
- Q5: What English language test scores are required for international graduate students?
- Q6: What documents and test requirements are needed for international graduate applicants?
- Q7: What documents are required for transfer students applying to undergraduate programs?
- Q8: What is required for first-year undergraduate applicants?
- Q9: What is the total cost of attendance for undergraduate students living on campus?
- Q10: How much is graduate tuition per credit hour?

5.2 Scoring Rubric

Each response was manually scored on a 1–5 scale considering accuracy (are the facts correct?), relevance (does it actually answer the question?), groundedness (does it stay within the retrieved context?), and clarity (is it well-written and concise?). A 5 means fully correct and clear. A 1 means the model produced nothing useful repeated the question, looped, or generated off-topic text. Scores of 3 or 4 reflect partial answers, usually missing one required fact.

5.3 Results

Table 1 shows the per-question scores for both models.

Table 1: Evaluation results - manual scores (1–5 scale)

#	Question	Gemma-2b-it	DistilGPT2	Winner
Q1	What is the priority deadline for international graduate students?	5.0	1.0	Gemma
Q2	What are the fall and spring application deadlines for graduate students?	4.5	1.0	Gemma
Q3	What documents are required for a graduate application?	4.0	1.0	Gemma
Q4	Do graduate applicants need GRE or GMAT scores?	4.0	1.0	Gemma
Q5	What English language test scores are required for international grad students?	4.0	1.0	Gemma
Q6	What documents/test requirements are needed for intl. graduate applicants?	3.5	1.0	Gemma
Q7	What documents are required for transfer undergrad applicants?	4.0	1.0	Gemma
Q8	What is required for first-year undergraduate applicants?	5.0	1.0	Gemma
Q9	What is the total cost of attendance for undergrad students living on campus?	3.0	1.0	Gemma
Q10	How much is graduate tuition per credit hour?	3.5	1.0	Gemma
	Average	**3.75**	**1.00**	**Gemma (10/10)**

Gemma-2b-it won all ten questions and averaged 3.75. It performed best on single-fact questions like Q1 and Q8 (both 5.0) and lower on questions requiring multi-chunk synthesis like Q9 (3.0) and Q6 (3.5). DistilGPT2 scored 1.0 on every question without exception.

6. Analysis and Discussion

6.1 Why DistilGPT2 Failed Completely

DistilGPT2 produced no useful answers across all ten questions. The core issue is that without instruction tuning it has no mechanism for following a structured prompt it simply predicts whatever token is most likely to come next given the input. In practice this meant it either echoed parts of the question back, continued the retrieved chunks as if writing more document text, or looped on a short fragment until hitting the 100-token limit. Wei et al. (2022) showed that instruction tuning is what enables models to generalize to structured tasks in zero-shot settings, and this experiment confirms that finding in a very direct way. Retrieval gives the model the right information, but if the model cannot interpret the instruction format, that information is wasted.

6.2 Retrieval Failures and How I Fixed Them

Early versions of the retriever failed on Q3, Q6, and Q8 because it kept returning undergraduate-related chunks for graduate-specific questions. The problem is that both file types use similar vocabulary words like application, documents, and requirements appear in both. Pure cosine similarity had no way to distinguish which source was relevant. Adding source-based boosting for the deadlines file and a penalty for off-topic spring admission chunks from non-deadline sources fixed this. After tuning, Q8 went from around a 2 to a 5. This was the most useful debugging I did on the whole project and it showed me that heuristic rules on top of semantic retrieval can be really valuable in small corpora where topic overlap is high.

6.3 Chunking and Incomplete Context

The worst failure case for Gemma was Q9, which received only a 3.0. Looking at what was retrieved, the relevant chunk contained the beginning of a cost-of-attendance sentence but the actual dollar figure had been cut off by the 300-character window. Gemma identified the right area of the document but could not finish the answer because the number was not there. Sentence-aware chunking splitting at sentence boundaries before applying a length limit would fix this. It is probably the single most impactful improvement I would make if continuing this project.

6.4 Model Comparison Summary

Gemma-2b-it worked well for the task overall. It followed the one-sentence format consistently, stayed within the retrieved context, and produced grammatically correct answers. Its weaknesses showed up on questions requiring synthesis across multiple chunks. DistilGPT2 is simply not suitable as a RAG generator without fine-tuning it treats the entire prompt as text to continue rather than as an instruction to follow. The performance gap between the two is not subtle; it is categorical.

7. Conclusion

UniGuide AI is a working RAG system for UNH admissions Q&A built using publicly available models and a small hand-collected corpus. The main findings are: Gemma-2b-it paired with dense retrieval can answer domain-specific admissions questions with reasonable accuracy, instruction tuning is essential and cannot be substituted by retrieval alone, and retriever engineering (source boosting, keyword penalties) matters as much as model choice in small corpora with topic overlap. The biggest remaining issue is character-level chunking, which occasionally gives the model incomplete context and leads to partial answers.

Future improvements would include sentence-aware chunking, a FAISS index for faster retrieval at scale, automated metrics like ROUGE or BERTScore for more objective evaluation, and a simple web interface so that prospective UNH students can actually use the system.

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